# Inventory Management User Guide

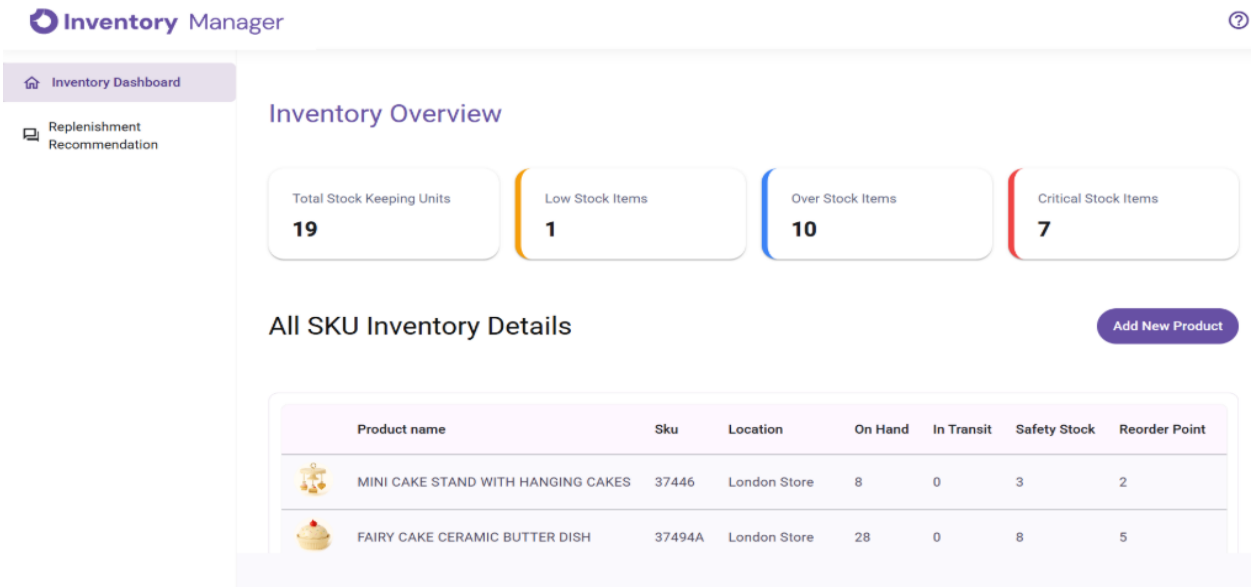
## Goal:

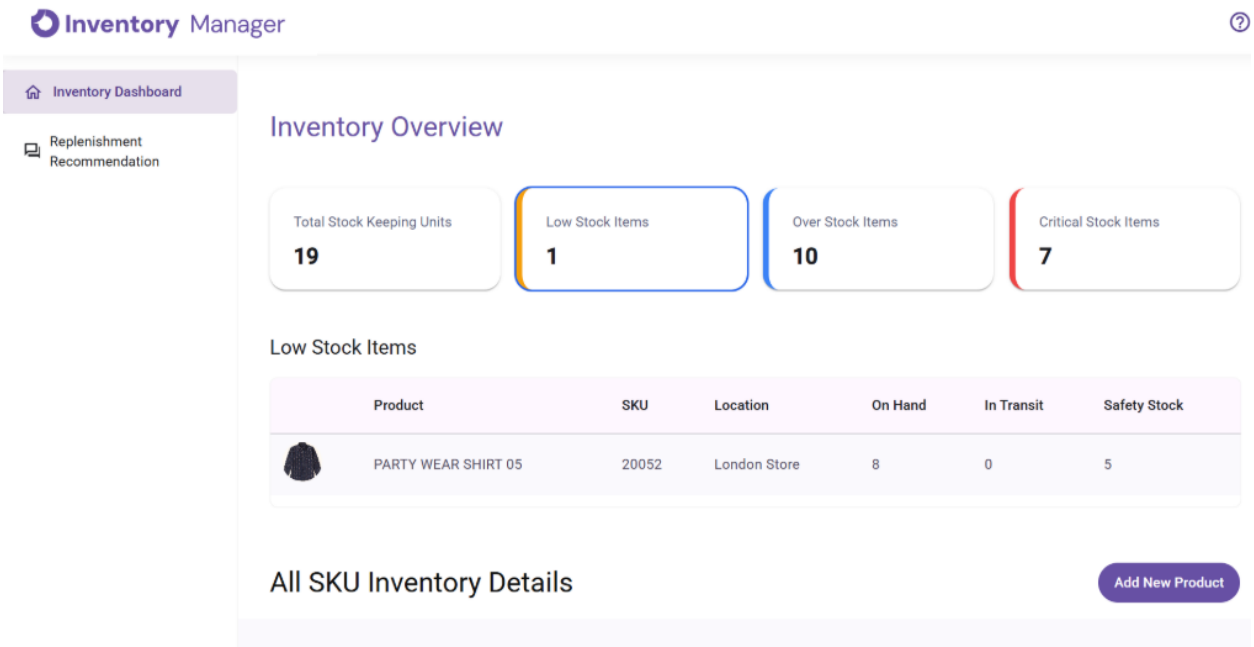
Perform real-time, **policy-compliant** inventory replenishment whenever **stock_levels.on_hand** falls below the **reorder point** by coordinating specialized agents **Demand**, **Inventory**, **Supplier**, and **Policy** through a central **Coordinator**. The system leverages **AlloyDB**, **AlloyDBVectorStore**, and **Vertex AI embeddings** to retrieve supplier details and policy documents with citations, calculate **recommended order quantities** (accounting for **MOQ**, **caps**, and **substitutions**), generate a **purchase order (PO) draft**, and deliver an approval summary.

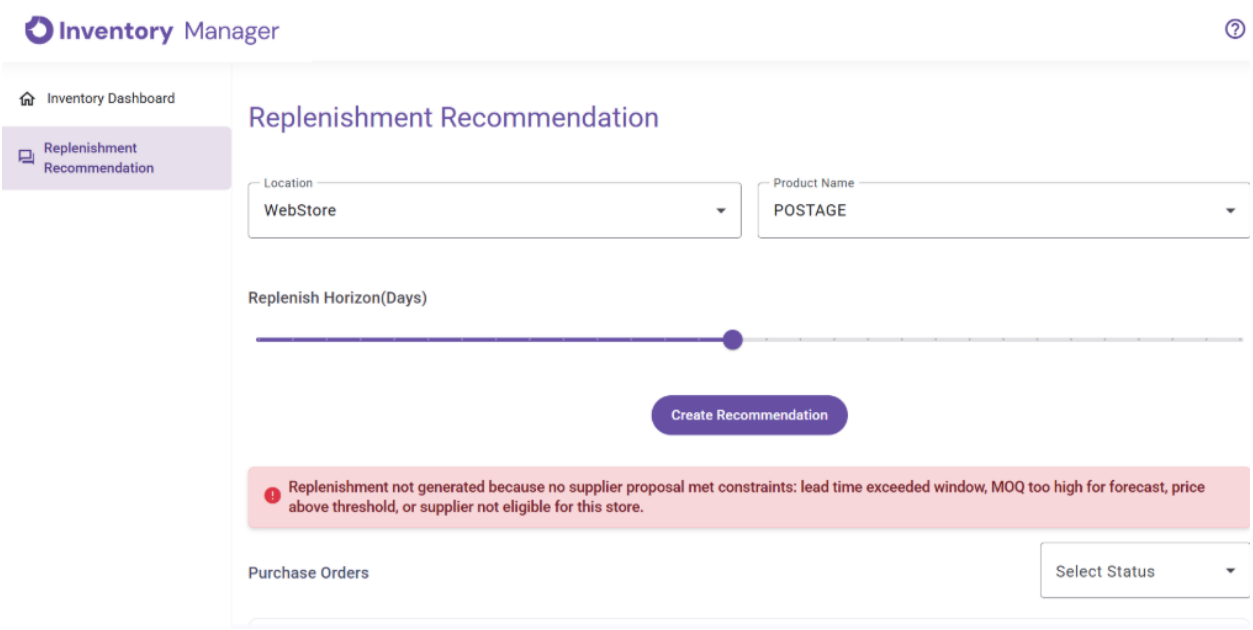
## Workflow:

1. **Inventory trigger** initiates the process.
2. A **Coordinator** prepares the end-to-end plan.
3. **Demand** is evaluated from mv_daily_demand.
4. **Supplier** data is pulled (MOQ, lead time, cost) along with **RAG** over retail documents.
5. A **Policy Agent** performs **RAG with citations** to validate rules.
6. The system computes required **quantity**, applies **caps/MOQ**, and finalizes the **supplier** selection.
7. A **PO draft** is generated and users are **notified** to approve, edit, or reject.
8. (Optional) Deploy via **Cloud Run**, with **Pub/Sub** and **Scheduler** for automation.

**Sample Images:**






**Key CloudSQL Components:**

- **CloudSQL for PostgreSQL** serves as the primary operational database, maintaining authoritative records for products, stores, stock levels, transactions, suppliers, and purchase orders.
- **AlloyDBVectorStore combined with VertexAIEmbeddings** enables RAG-based retrieval over documents such as supplier notes, seasonal memos, policies, and SOPs, returning results with citations.
  *Note: VECTOR(d) columns must match the embedding model dimension — d = 768 for Vertex textembedding005.*
- **ScaNN Index** is the recommended vector-search option at scale, configured with USING scann (...), tuned num_leaves, and optionally max_num_levels = 2, followed by an ANALYZE operation for optimization.
- **pgvector (Fallback Option):** Supports HNSW or IVFFlat indexing using the appropriate operator class (e.g., vector_cosine_ops) and the matching <=> similarity operator.
- **LangChain Integrations:**
  - AlloyDBEngine for secure, IAM-authenticated connection pooling
  - AlloyDBVectorStore + VertexAIEmbeddings for retrieval workflows
  - *(Optional)* SQLDatabaseToolkit for structured SQL interactions

- **Google Cloud Storage (GCS)** acts as the centralized repository for retail.docs, including policies, SOPs, seasonal notes, and ingestion artifacts.
- **Orchestration & Runtime Components** *(optional for production)*:
    - **Cloud Run** for hosting agent services and workflow endpoints
    - **Secret Manager** for securely storing non-IAM secrets

## Technical Challenge & Business Impact:

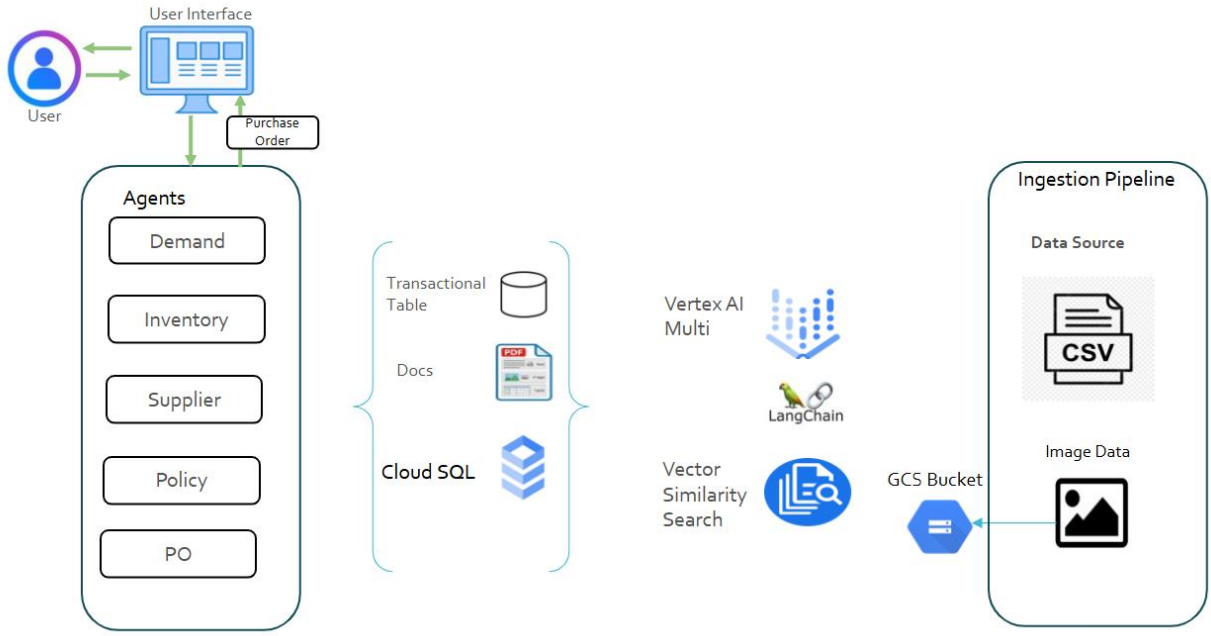
**Challenge:**

Manual, spreadsheet-based replenishment cannot keep up with rapid **fluctuations in demand,** supplier constraints (MOQ, lead time, cost), or **policy requirements** (caps, substitutions) at scale. Critical operational knowledge is often **scattered across documents**, resulting in slow, inconsistent decisions and limited **auditability**. Traditional systems also struggle to combine **real-time transactional data** with **policy-aware logic** and **explainable document retrieval**, leading to inefficiencies and missed opportunities.

**Impact:**

A real-time, **agent-driven**, **RAG-enabled** replenishment workflow built on **AlloyDB** and **GCP** automates **policy-compliant** purchase order generation with full **citation traceability**. This approach reduces **stockouts** and **excess inventory**, shortens the **decision-to-order cycle**, and improves overall **operational precision**. **Demand-based ordering**, **supplier-aware optimization** (MOQ, lead time, cost), and **policy-bounded recommendations** (caps, substitutions) together deliver higher **service levels**, improved **inventory turns**, and lower carrying costs while ensuring **audit-ready transparency** and **production-grade robustness**.

**What the "Single Pass" Does:** *(as implemented in src/workflows/run_single_pass.py)*

- Selects a frequent **(store, SKU)** pair from recent transactions and verifies that the required stock_levels and supplier mappings exist.
- Invokes the **Coordinator**, which executes the Demand, Inventory, Supplier, and Policy (RAG) agents using AlloyDBVectorStore and VertexAIEmbeddings.
- Calculates recommended_qty = max (0, (forecast × horizon + safety) – net_position), applies MOQ rounding, enforces policy caps, and determines the optimal supplier.
- Generates a purchase order draft, produces an approval summary (auto-approved in the demo workflow), writes the PO record, and outputs the generated **po_id**.

# Architecture Diagram